

Education

- 2016–2021 **Ph.D.**, *Computer Science*, Massachusetts Institute of Technology (Expected).
Advisor: Prof. Armando Solar-Lezama
Thesis: Neurosymbolic Learning for Robust Intelligent Systems.
- 2015–2016 **Master of Engineering**, *Computer Science*, Massachusetts Institute of Technology.
GPA: 5/5
Advisor: Prof. Armando Solar-Lezama
Thesis: Synthesis of Domain Specific CNF Encoders for Bit-Vector Solvers
- 2012–2016 **Bachelor of Science**, *Electrical Engineering and Computer Science*, Massachusetts Institute of Technology.
GPA: 4.9/5

Research focus

Artificial Intelligence and Program Synthesis. Developing *neurosymbolic* approaches for learning *program* models that are interpretable and generalizable, and applying them to several applications in *robotics*.

Experience

- Summer 2017 **Research Intern**, TOYOTA RESEARCH INSTITUTE, Cambridge, MA.
Worked on modeling human driving behavior using traffic data so that the models can then be used to synthesize/verify safe autonomous controllers.
- Summer 2016 **Research Intern**, MICROSOFT RESEARCH, Redmond, WA.
Developed a programming-by-example system to integrate the relational data from spreadsheets with the semi-structured web-data.
- Summer 2014 **Software Engineering Intern**, GOOGLE INC., Mountain View, CA.
Designed and implemented an android app for Content ID feature in YouTube.

Publications

- Under submission Jeevana Priya Inala, Jason Ma, Osbert Bastani, Xin Zhang, Armando Solar-Lezama.
Safe Human-Interactive Control Modulo Fault.
- NeurIPS 2020 Jeevana Priya Inala*, Yichen Yang*, James Paulos, Yewen Pu, Osbert Bastani, Vijay Kumar, Martin Rinard, Armando Solar-Lezama.
Neurosymbolic Transformers for Multi-Agent Communication.
- ICLR 2020 Jeevana Priya Inala, Osbert Bastani, Zenna Tavares, Armando Solar-Lezama.
Synthesizing Programmatic Policies that Inductively Generalize.
- ICRA 2019 Thais Campos*, Jeevana Priya Inala*, Armando Solar-Lezama, Hadas Krez-Gazit. (* equal contribution)
Task-based Design of Modular Ad-hoc Manipulators.
- SIGGRAPH ASIA 2018 Tao Du, Jeevana Priya Inala, Yewen Pu, Andrew Spielberg, Adriana Schulz, Daniela Rus, Armando Solar-Lezama, Wojciech Matusik.
InverseCAD: Automatic Conversion of 3D Models to CSG Trees.
- POPL 2018 Jeevana Priya Inala, Rishabh Singh.
WebRelate: Joining Web Data with Relational Data using Examples.
- TACAS 2017 Jeevana Priya Inala, Nadia Polikarpova, Xiaokang Qiu, Ben Lerner, Armando Solar-Lezama.
Synthesis of Recursive ADT Transformations from Reusable Templates.
- EuroSys 2016 Nathaniel Herman, Jeevana Priya Inala, Yihe Huang, Lily Tsai, Eddie Kohler, Barbara Liskov, Liuba Shrira.
Type-Aware Transactions for Faster Concurrent Code.

SAT 2016 Jeevana Priya Inala, Rohit Singh, Armando Solar-Lezama.
Synthesis of Domain Specific CNF Encoders for Bit-Vector Solvers.

Awards

- 2016 - 17 Microsoft Research Women's Fellowship
- 2016 Charles and Jennifer Johnson MEng Thesis First Place Award
- 2016 First place in ACM student research competition grand finals
- 2015 First place in PLDI student research competition
- 2014 - 15 Actifio Undergraduate Research and Innovation Scholar
- 2012 Gold medal at 13th Asian Physics Olympiad, India
- 2011 Gold medal and Best in Theory in 5th International Olympiad in Astronomy and Astrophysics, Poland
- 2012 Silver medal and Asian Girl topper in 43rd International Physics Olympiad, Estonia
- 2012 Secured rank 21 in the Indian Institute of Technology (IIT) Joint Entrance Examination

Skills

- Machine Learning Reinforcement Learning; Multi-Agent RL; Transformers; Generative Adversarial Network (GAN); Variational Autoencoder (VAE);
- Program Synthesis Sketch synthesizer; SMT solvers like Z3, CVC4; Version Space Algebra;
- Languages Python, PyTorch, Java, C++

Talks/Posters

- Dec 2020 **NeurIPS conference.** Neurosymbolic Transformers for Multi-Agent Communication.
- April 2020 **ICLR conference.** Synthesizing Programmatic Policies that Inductively Generalize.
- Feb 2020 **AAAI GenPlan workshop.** Synthesizing Programmatic Policies that Inductively Generalize.
- Jan 2018 **POPL conference.** WebRelate: Joining Web Data with Relational Data using Examples.
- Apr 2017 **TACAS conference.** Synthesis of Recursive ADT Transformations from Reusable Templates.
- Aug 2016 **Microsoft Research.** WebRelate: Joining Web Data with Relational Data using Examples.
- Jul 2016 **SAT conference.** Synthesis of Domain Specific CNF Encoders for Bit-Vector Solvers.
- Jul 2016 **SMT workshop.** Synthesis of Domain Specific CNF Encoders for Bit-Vector Solvers.
- Jun 2016 **ExCAPE PI meeting.** Synthesis of Domain Specific CNF Encoders for Bit-Vector Solvers.
- Jun 2015 **PLDI student research competition.** Synthesis of Recursive ADT Transformations from Reusable Templates.

Reviewing

Reviewer for artifact evaluation for CAV 2020 and POPL 2019.
Reviewer for journal papers in Robotica.